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Palaeomagnetically inferred tectonic rotations of the Abruzzi and northwestern Umbria

Péter Márton [a] and Myriam D'Andrea [b]

[a] *Geophysics Department, Eötvös Loránd University, Ludovika tér 2, H-1083 Budapest, Hungary*
[b] *Italian Geological Survey, Largo S. Susanna 13, I-00187 Rome, Italy*

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ABSTRACT

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Palaeomagnetic results are presented from the Abruzzi and northwestern Umbria (Monte Nerone area) for the Cretaceous and the combined Jurassic and Cretaceous timespans. The results for the Cretaceous, when combined with previously published work, indicate that both the Abruzzi and northwestern Umbria underwent a post-Cretaceous tectonic rotation relative to Apulia. The rotation of the Abruzzi is 40° clockwise while that of northwestern Umbria is 22–23° counterclockwise. Apulia itself has been rotated only 11° relative to Africa according to a recently proposed apparent polar wander path for Africa.

Introduction

Geological evidence suggests that the Apennine mountain range consists of several faulted and thrust geological units. These units originally belonged to the various isopic zones of the Alpine geosyncline in the southern margin of the Mesozoic Tethys (Consiglio Nazionale delle Richerche, 1981). In the northern and central Apennines the external zones, which now trend north–south, are found. Each of these zones consists of tectonostratigraphic units derived from the deformation of pre-existing palaeogeographic units and attained their present position following several tectonic phases during the Neogene. In the west, the Tuscan zone (emplaced in Late Oligocene–Early Eocene) was thrust over the Umbria-Marche zone by the end of the Miocene. Later (Messinian–Pliocene) Umbria-Marche was itself thrust over the Abruzzi-Latium sequence to the east. Important easterly displacements of the Abruzzi-Campania zone over terrigenous Miocene beds have also been demonstrated. The Apulian zone, however, which forms the lowest structural unit, is considered to have been part of the stable foreland of Adria during the entire geosynclinal period (Celet, 1977).

Palaeomagnetic techniques have been used to measure the megatectonic movement of the Adriatic continental margin as a whole relative to Africa, as well as to gain insight into the tectonic rotations of the different tectonic units, on both a local and regional scale (see Lowrie, 1986, for a review). The present work is concerned with the Abruzzo and northwestern Umbrian regions for which we obtained new palaeomagnetic results. We will use these, in conjunction with previous data, to show the presence of significant tectonic rotations of both sampling areas.

Sampling and laboratory treatment

Individually oriented hand samples were collected, which were subsequently drilled into standard sized specimen cores (2.5 × 2.2 cm) in the laboratory. Each sampling site represents a stratigraphic section of a thickness ranging from a few metres to several tens of metres with the same

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bedding attitude. In the Abruzzi, pelagic limestones of Cretaceous age were sampled east of the Profluo-Tasso-Sagittario line (Colacicchi and Praturlon, 1965). A total of 135 samples were collected from 4 Early Cretaceous, 1 Albian and 4 Turonian–Santonian sites. The Early Cretaceous sites are located along the northern slope of the Serra Sparvera (sites A1 and A2, Fig. 1a), at the southern limb of the Serra del Feudo (A3) and near Lago Pantaniello (A4), respectively. The Turonian–Santonian sites are located along the southern side of the Serra Sparvera (A5–A8), part of which is a reversed thrust with a steep southerly dip (A7 and A8). Finally, the Albian site is a few hundred metres above Lago Pantaniello along the eastern side of Serra Rocca Chiarano (A9) (Fig. 1a).

In Umbria, pelagic limestones were sampled from the Presale valley section along tourist path N.13 of the Monte Nerone area near Piobbico. A

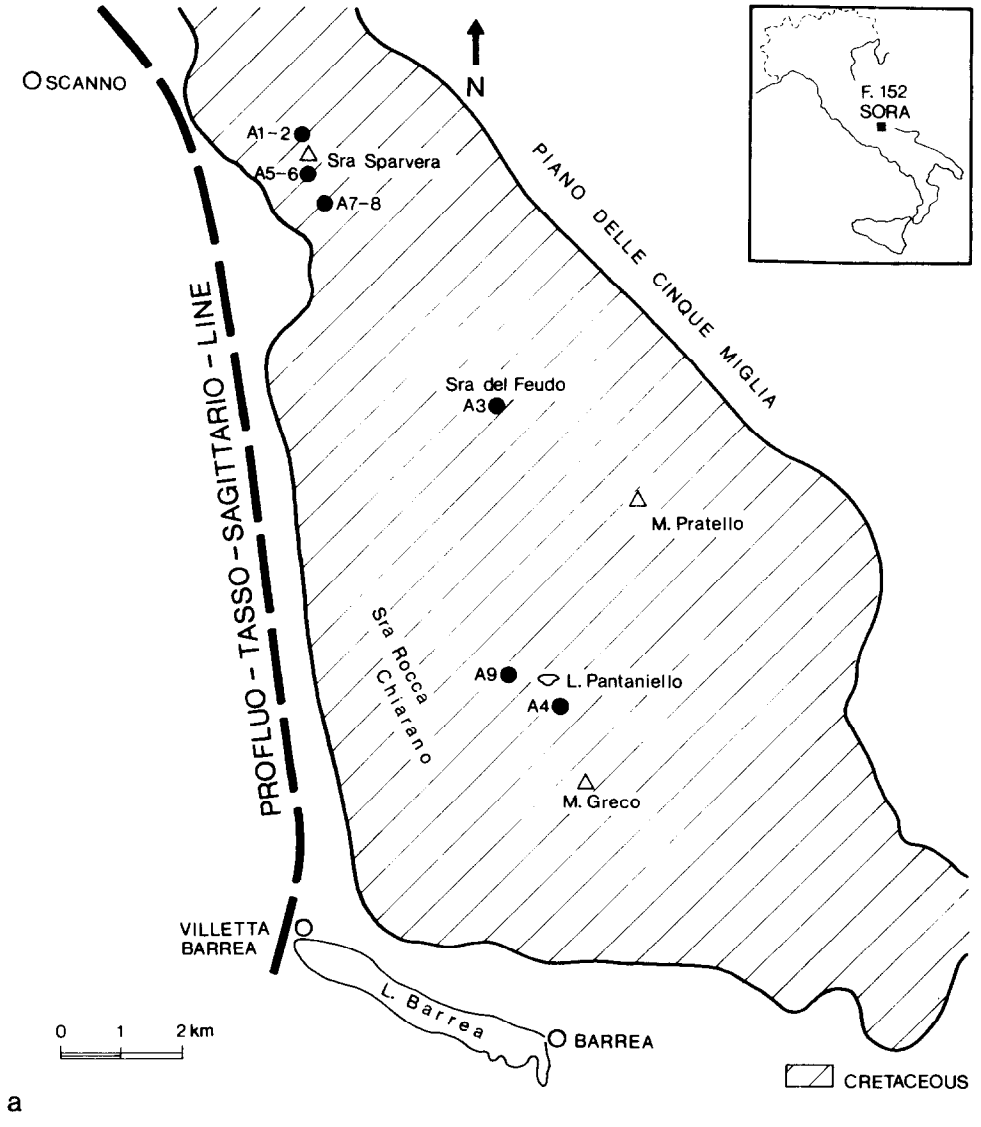


Fig. 1. Location of the areas studied. (a) The Abruzzi. (b) Northwestern Umbria. (After Italian Geological Survey maps shown in the insets). A1–A9 and U1–U6 = sampling sites for the present study. The mean palaeomagnetic directions for northwestern Umbria (a) were determined by averaging the palaeomagnetic results of this paper and earlier studies from Bosso, Presale, Gorgo a Cerbara (Hirt and Lowrie, 1988) and F. Giordano (Cirilli et al., 1984) for the Early Cretaceous and IFM, IRO (VandenBerg et al., 1978) and Poggio Guiane (Lowrie et al., 1980b) for the Albian-Santonian (see Table 3).

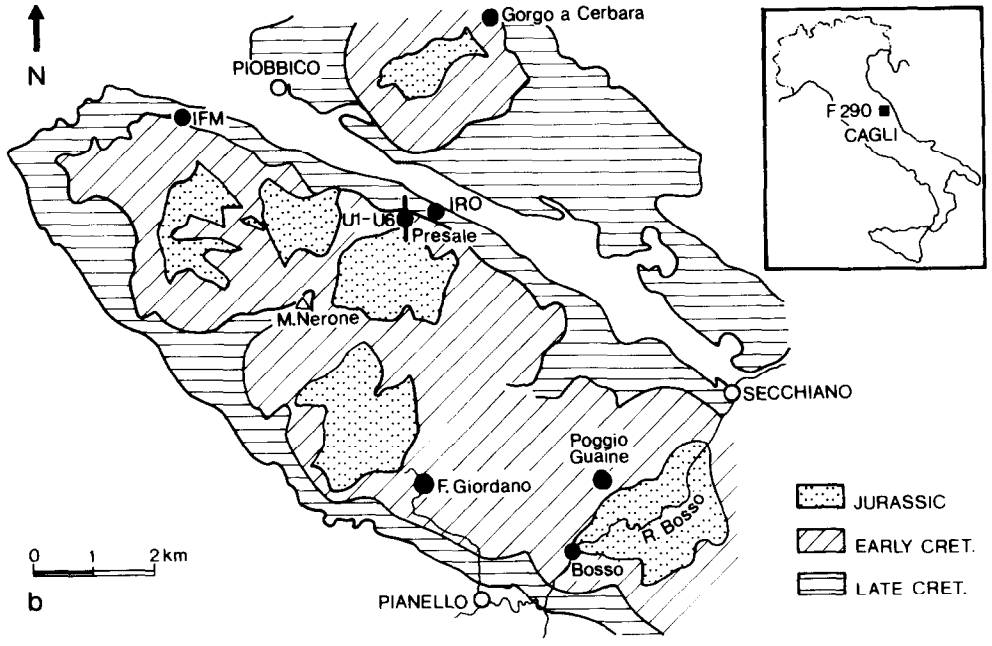


Fig. 1 (continued).

total of 105 hand samples were collected from six sites (U1–U6) which represent the following geological stages: Sinemurian–Pliensbachian (12 samples), Toarcian–Bajocian (18), Kimmeridgian–Tithonian (9), Neocomian–Barremian (24), Albian–Cenomanian (16) and Turonian–Santonian (26) (Fig. 1b).

The magnetic measurements were carried out on a Jelinek (JR-4) magnetometer with a root mean square noise level of 4×10^{-6} A/m. The specimens were demagnetized stepwise in a Schonstedt oven as far as possible while the susceptibility was monitored after each demagnetization step with a KLY-1 bridge. The natural remanent magnetization (NRM) components were identified from orthogonal demagnetization (Zijderveld) plots which were constructed and evaluated for each specimen. Representative examples of typical demagnetization behaviour are shown in Figure 2.

The magnetic mineralogy of the samples was inferred by experiments on induced remanent magnetization (IRM) acquisition and demagnetization. Selected specimens from each site were subjected to incremental magnetization at room temperature in fields up to 0.95 T. Most of the resulting IRM acquisition curves are dominated by the presence of a magnetic mineral which is saturated in relatively low fields ($h < 0.15$ T) and which, by analogy to studies of pelagic limestones (e.g., Lowrie and Heller, 1982), we interpret to be magnetite (upper diagram Fig. 3a). In order to verify this interpretation, and to help identify the other magnetic minerals whose presence is indicated on some IRM curves by contributions acquired in fields greater than 0.15 T, an inverse IRM was given to each specimen subjected to these studies, which was subsequently demagnetized thermally in suitable steps up to 680°C. (The inverse IRM was composed of an IRM induced along the specimen's $-z$ axis in a 0.95 T field and a second IRM of the same polarity applied along the $+z$ axis in a 0.15 T field.) Prior to demagnetization, the majority of the test samples had a net IRM in the direction of the low field component. The high field component (if any) became discernible only after demagnetization at a temperature near the unblocking temperature of the magnetite (585°C) and was eventually reduced to zero at 680°C (Figs. 3a and b). Obviously, the low field component is carried by magnetite and the high field component is carried by haematite. However, the importance of the haematite as a contributor to the IRM is

generally much less than that of the magnetite, except for a few horizons which contain more haematite and/or exhibit a more complex mineralogy (Figs. 3c and d). For example, Figure 3c shows a dominant haematite fraction, an important magnetite phase and perhaps some goethite, as indicated by the slight rise in IRM at the beginning of the demagnetization diagram. In

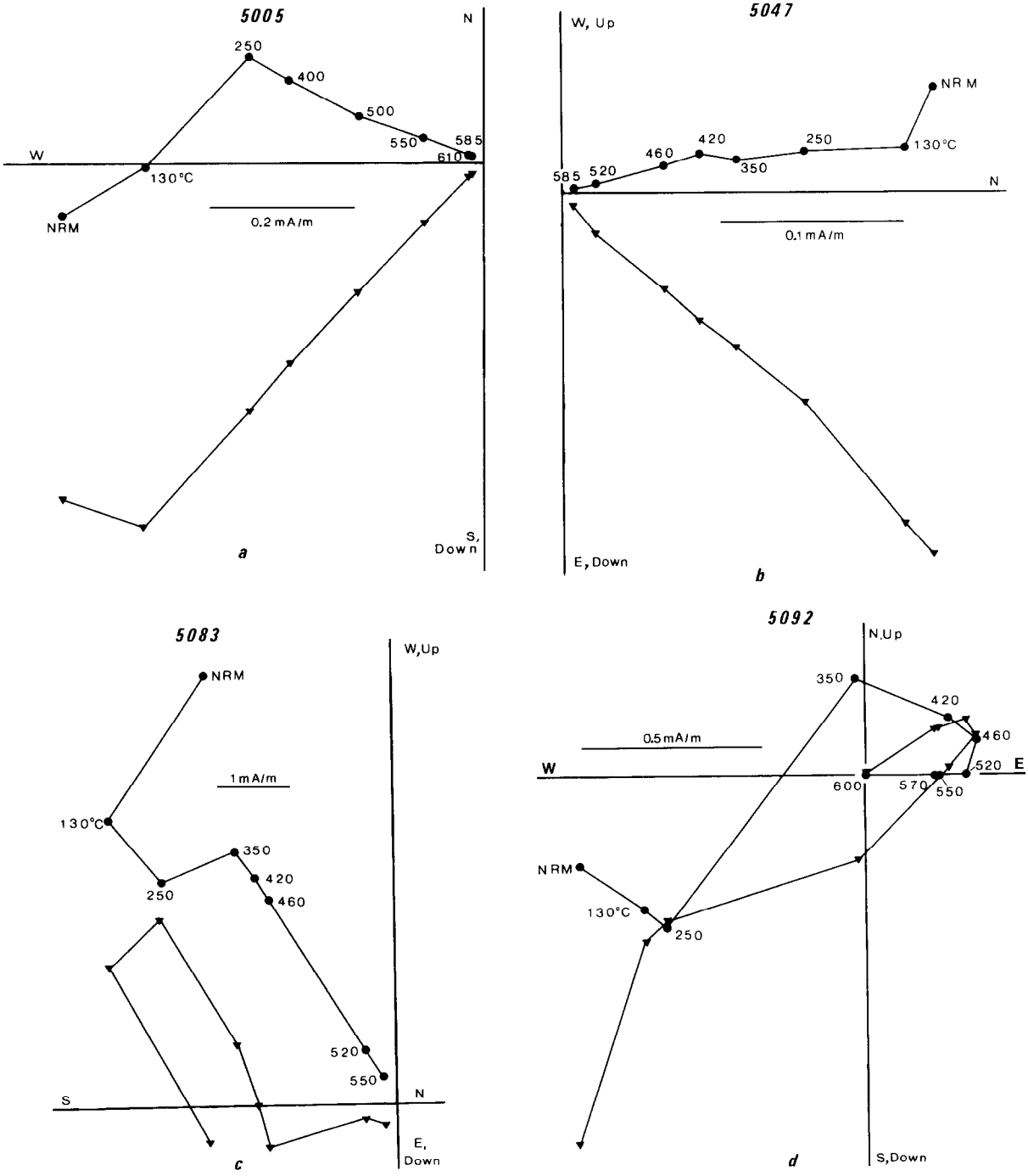


Fig. 2. Typical demagnetization plots of pelagic limestone samples from the Abruzzi—(a) A7 (b) A4—and the Presale section in Umbria—(c) U2, (d) U3, (e) U6. Endpoints of the magnetization vectors are shown after correction for the tectonic tilt. Dots = endpoints projected onto horizontal planes; triangles = endpoints projected onto vertical planes.

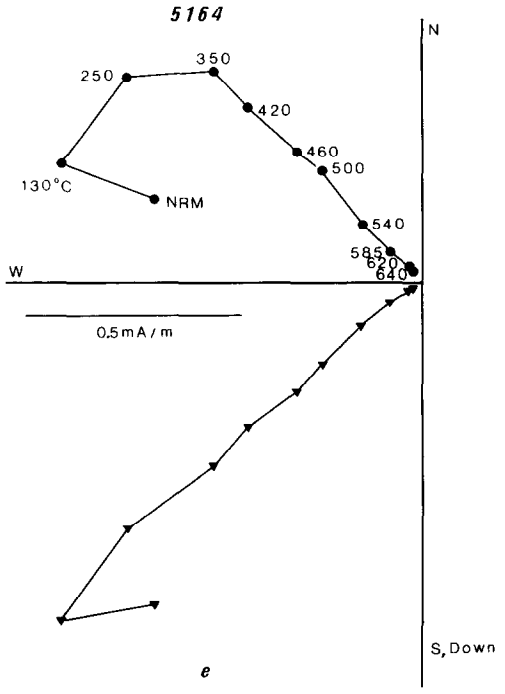


Fig. 2 (continued).

addition to the magnetite and haematite phases, Figure 3d shows a third component, with an IRM oriented in the high field direction, which unblocks completely at 400°C. This component is probably a second generation haematite with a grain size distribution typical of pigments.

Discussion of palaeomagnetic results

Abruzzi

Table 1 contains our palaeomagnetic results from the Abruzzo region and Jackson's result from Monte Maiella, which is located some 30 km NNE of our sampling area (Jackson, 1990). One of the Berriasian sites at Serra Sparvera (A1) yielded no palaeomagnetic result owing to the extremely weak magnetization of these limestones. Similarly, the majority of the samples from the Serra del Feudo and Lago Pantaniello sites (A3 and A4) were too weak for precise measurement, so the data were derived from a much smaller number of samples than we had collected at these sites (see also Jackson, 1990).

The fold test (McFadden and Lowes, 1981) between the four Early Cretaceous sites (Table 1) is significant in the sense that the hypothesis of a common true mean direction for any combination of two sites may not be rejected at the 95% level after unfolding. Consequently, it may be accepted that the observed magnetization was acquired before distortion of the beds (McFadden and Jones, 1981). The evidence of magnetite (Fig. 3a) and the existence of both magnetic polarities suggest an early acquisition, which we assume to be contemporaneous with the stratigraphic age (Early Cretaceous).

Three of the younger Cretaceous sites (Albian–Santonian) gave unfolded directions which, when tested statistically (McFadden and Lowes, 1981), cannot be distinguished from each other at the 95% confidence level. However, the fourth direction (Table 1, A7–A8) is different in that its declination is rotated 65° counterclockwise relative to the mean declination of the former three directions. Since these limestones predominantly contain magnetite, the magnetization determined for each site is probably "primary", so that the anomalous declination of the reverse thrust block is very probably the consequence of a local rotation during final emplacement.

Northwestern Umbria, Monte Nerone area

Common to all three sites of the Jurassic limestones in the Presale section is a heavy overprint which dominates the initial magnetization of the samples. The bulk of the overprint was removed by thermal treatment and the demagnetization curves showed a single characteristic component, usually after 420°C (as in Figs. 2c and d), and only a few samples had to be rejected for not responding to thermal treatment. In order to reduce the effect of the residual overprint after the cleaning procedure, the normal and reversed samples were separated at each site and the site mean direction of magnetization (Table 2) was computed as the average direction of the mean normal and reversed directions (Irving, 1964).

The Maiolica limestones (Hauterivian–Barremian) also showed alternating normal and reversed polarity zones, whereas the Scaglia (Al-

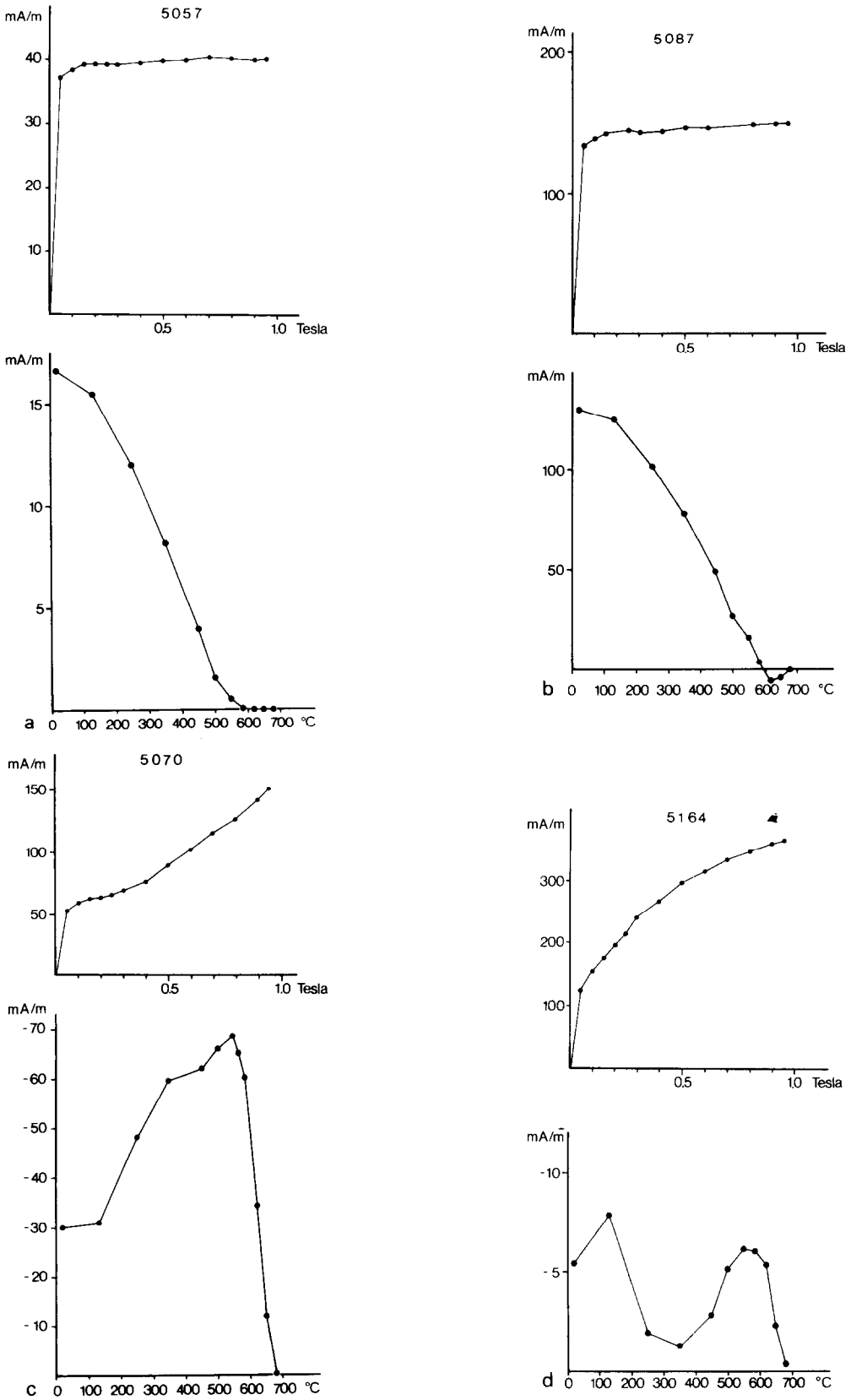


Fig. 3. IRM acquisition (upper diagram) and demagnetization behaviour of inverse IRM (lower diagram) for a selection of samples (see text for explanation). (a) Samples with magnetite. (b) Samples with predominant magnetite plus some haematite. (c) and (d) are examples of more complex mineralogy (see text).

bian–Cenomanian and Turonian–Santonian) yielded only normal polarity samples, all in agreement with the known polarity time scale for the Cretaceous (e.g., Lowrie et al., 1980a). Half of the rejections had to be made for extremely weak magnetizations (mostly Maiolica samples) and the other half came from samples from two zones in the Turonian–Santonian subsection, which probably possess a secondary magnetization. This secondary magnetization is of anomalous direction—$D = 302°$, $I = 52.6°$, $k = 37.2$, $\alpha_{95} = 9.2°$ before, and $D = 348.8°$, $I = 21.8°$ after tectonic tilt correction (k and α apply to both cases)—and was probably imprinted much later (possibly during the Neogene) in or close to the present position of the beds, but before the tectonic rotation of the region. The palaeomagnetic results for the three Cretaceous sites also are included in Table 2.

The dominant magnetic mineral of the Presale valley limestones is the magnetite (Fig. 3b), so it may be identified as the main carrier of the characteristic magnetizations listed in Table 2. Haematite may also contribute to the characteristic remanence of some horizons, but when its contribution becomes discernible, during demagnetization above the unblocking temperature of the magnetite, it is of the same direction as that

TABLE 1

Summary of palaeomagnetic results from the Abruzzi ($\varphi = 41.9°$, $\lambda = 14.0°$)

Sampling site (age interval)	N/N_0	D^0 D_c^0	I^0 I_c^0	k	α_{95}^0	Treatment (°C)
Early Cretaceous sites:						
Serra Sparvera (A2)	9/24	327.8	21.7	66	6.4	250–500
(134–128 My)		329.1	23.3	66	6.4	
Serra del Feudo (A3)	7/21	293.1	65.6	47	8.9	300–420
(134–117 My)		334.8	45.4	47	8.9	
Lago Pantaniello (A4)	9/51	350.9	37.5	33	9.0	300–510
(134–128 My)		345.6	32.5	33	9.0	
Monte Maiella [1]	9/46	314.8	4.3	44	–	af demagnetization
(140–113 My)		326.0	42.1	44	–	
Mean		324.9	33.4	7	–	
(number of sites = 4)		334.0	36.1	44	14.0	
Albian–Santonian sites:						
Serra R. Chiarano (A9)	11/12	18.7	27.2	79	5.2	450–565
(108–92 My)		9.0	36.4	79	5.2	
Serra Sparvera (A5)	7/7	29.8	43.4	80	6.8	$\geqslant 500$
(92–84 My)		10.9	46.0	80	6.8	
Serra Sparvera (A6)	6/6	17.5	40.3	32	12.1	> 500
(92–84 My)		10.5	40.1	32	12.1	
Mean		21.7	37.1	65	15.4	
(number of sites = 3)		10.1	40.8	273	7.5	
Serra Sparvera (A7–A8)	8/14	327.6	1.4	27	10.8	250–550
		305.2	36.3	55	7.6	

[1] This result is from Jackson (1990).

N/N_0 is the number of useful samples/samples collected.

D^0 and I^0 are the mean palaeomagnetic declination and inclination before tectonic tilt correction.

D_c^0 and I_c^0 are those after tectonic correction.

k is the precision parameter.

α_{95} is the radius of the circle of confidence (Fisher, 1953).

The last column shows the temperature range in which the characteristic magnetization was defined.

Numerical ages are based on the Haq et al. (1987) time scale.

of the magnetite (Fig. 2e). Thus, we may conclude here by suggesting that the magnetic directions for the Presale sequence (Table 2) are characteristic of the related stratigraphic ages.

Tectonic rotations

The tectonic rotations within the Italian peninsula may be referred to the stable Apulian foreland which itself underwent a "microplate" rotation relative to Africa during the Tertiary. The current estimate of this rotation is 17° (Lowrie, 1986), that is, 8° smaller than that given by VandenBerg (1983). Since some new data have recently become available both from Apulia and Africa, it is reasonable first to address the question of the magnitude of this rotation.

The Apulian palaeomagnetic data may be subdivided into an early Cretaceous and a late Cretaceous group, in the same way as those from the Abruzzi and Umbria (Table 3). In selecting the reference data for Africa (Table 3) we followed Hargraves (1989), except that we rejected the Mlanje pole which was calculated from results obtained from only eight samples (see Hargraves, 1989). Figure 4 shows that the best fit between the Apulian and related African palaeomagnetic poles can be achieved by an 11° rotation, which means that the counterclockwise rotation of Apulia is probably smaller than the current estimate of it.

In what follows, we will assume that the northern and central Apennines shared the movements of the stable foreland during the late Mesozoic until they underwent an 11° tectonic rotation relative to Africa during the Tertiary. If this is correct, then both local and regional tectonic rotations in Umbria and the Abruzzi may be inferred from the differences between their palaeomagnetic poles and the related Apulian poles.

It has been shown that Umbria-Marche as a

TABLE 2

Palaeomagnetic results from the Presale valley section, Monte Nerone area (NW Umbria) ($\varphi = 43.6°$, $\lambda = 12.5°$)

Sampled geological stages (age interval)	N/N_0	D^0 D_c^0	I^0 I_c^0	k	α_{95}^0	Temperature (°C)
Sinemurian–Pliensbachian (U1) (201–186 My)	12/12	252.0 289.3	30.4 54.0	21.0	9.6	420–550
Toarcian–Bajocian (U2) (186–165 My)	14/18	260.9 310.1	40.0 50.8	57.0	5.3	420–550
Kimmeridgian–Mid-Tithonian (U3) (145–137 My)	7/7	254.8 268.8	5.9 33.6	73.0	7.1	(420)–550
Hauterivian–Barremian (U4) (121–113 My)	14/26	258.3 279.5	11.5 40.0	31.0	7.3	(350)–460
Albian–Cenomanian (U5) (108–92 My)	14/16	264.6 302.4	−7.1 43.7	26.1 32.0	7.9 7.2	460
Turonian–Santonian (U6) (92–84 My)	16/26	264.4 302.3	17.2 37.4	22.0 32.0	8.1 6.6	420–585

Symbols as in Table 1.

whole has not been rotated appreciably relative to Apulia (Lowrie, 1986; Hirt and Lowrie, 1988), though a regional N–S difference in the declination data has been noted and interpreted as being probably due to uncompensated tectonic effects (Hirt and Lowrie, 1988). However, the Cretaceous palaeomagnetic data from the Monte Nerone area (Fig. 1b) are internally consistent (Table 3) and indicate a 22–23° counterclockwise rotation (about the local vertical) relative to Apu-

TABLE 3

Summary of Cretaceous palaeomagnetic poles

Geological stages (age interval)	$N/(n)$	ϕ^0	Λ^0	k	α_{95}^0	Reference
Africa						
Tihonian–Barremian (142–113 My)	10	47.6	269.9	26	9.7	1
Berriasian–Barremian (131–113 My)	40	48.3	266.6	53	3.1	1
Aptian–early Albian (113–109 My)	7	61.8	259.0	339	4.0	1
Late Albian–Santonian (100–86 My)	15	69.3	258.1	21	8.5	1
Late Albian–Santonian (101–83 My)	14	64.1	226.1	59	5.2	1
Turonian (90–? My)	(170)	69.1	240.0	26	4.9	1
Turonian–Maastrichtian (92–63 My)	31	64.6	258.1	32	4.6	1
Campanian (< 83 My)	12	68.8	220.8	(99)	(4.4)	1
Maastrichtian (75–? My)	(211)	63.5	219.6	42	4.1	1
Apulia						
Early Cretaceous (134–96 My)	11	38.0	278.0	–	5.0	2, 3
Cenomanian–Santonian (96–84 My)	29	57.7	257.5	312	7.0	2, 4, 5
NW Umbria						
Early Cretaceous (140–113 My)	5	21.6	292.0	154	6.2	6, 7, 8
Albian–Santonian (108–84 My)	4	39.7	275.8	230	6.1	7, 8, 9, 10
Abruzzi						
Early Cretaceous (140–113 My)	4	59.3	247.7	60	12.0	8, 11
Albian–Santonian (108–84 My)	3	69.7	166.3	469	5.7	8

References: 1 = Hargraves (1989); 2 = VandeBerg (1983); 3 = Lowrie (1986); 4 = Márton and Nardi (1990); 5 = Channell (1977); 6 = Cirilli et al. (1983); 7 = Hirt and Lowrie (1988); 8 = this paper; 9 = Lowrie et al. (1980b); 10 = VandenBerg et al. (1978); 11 = Jackson (1990).

$N/(n)$ is the number of sites/(samples).

ϕ and Λ are the geographic latitude and longitude of the palaeomagnetic pole.

k and α_{95}^0 are Fisher statistical parameters (Fisher, 1953).

Numerical ages for the Italian poles are based on the Haq et al. (1987) time scale.

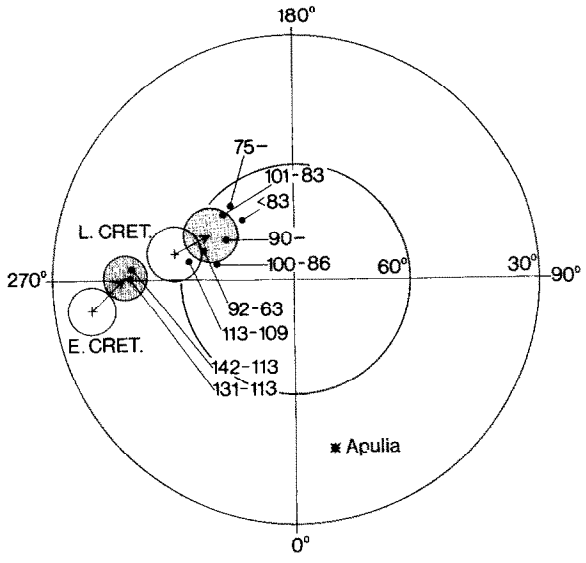


Fig. 4. Comparison of Cretaceous palaeomagnetic poles for Apulia and Africa, stereographic projection. The Apulian poles (unshaded) are shown to have been rotated 11° counterclockwise relative to their best fit position (shaded) to the African apparent polar wander curve—dots (with time marks in My). Asterisk shows the geographic position of Apulia.

lia, as shown by Figure 5. It appears, therefore, that the northwestern tip of Umbria moved differently from the rest of the region during the late Tertiary orogeny.

Figure 5 also shows that the palaeomagnetic poles for the Abruzzi (Table 3) are rotated 40° clockwise relative to their Apulian counterparts. A somewhat smaller rotation (24° ± 15°) in the same direction has been suggested by Jackson for the Maiella platform from a single Early Cretaceous result. (Monte Maiella is located some 30 km NNE of our sampling area in the Abruzzi.) The present data, including Jackson's result, however, are indicative of a greater rotation of not only a thrust (Maiella) but perhaps the entire Abruzzo region. Obviously, further work is needed to substantiate this suggestion.

In summary, we may conclude that the northwestern part of Umbria (Monte Nerone area) was rotated 22–23° counterclockwise and the Abruzzi was rotated 40° clockwise, both with respect to

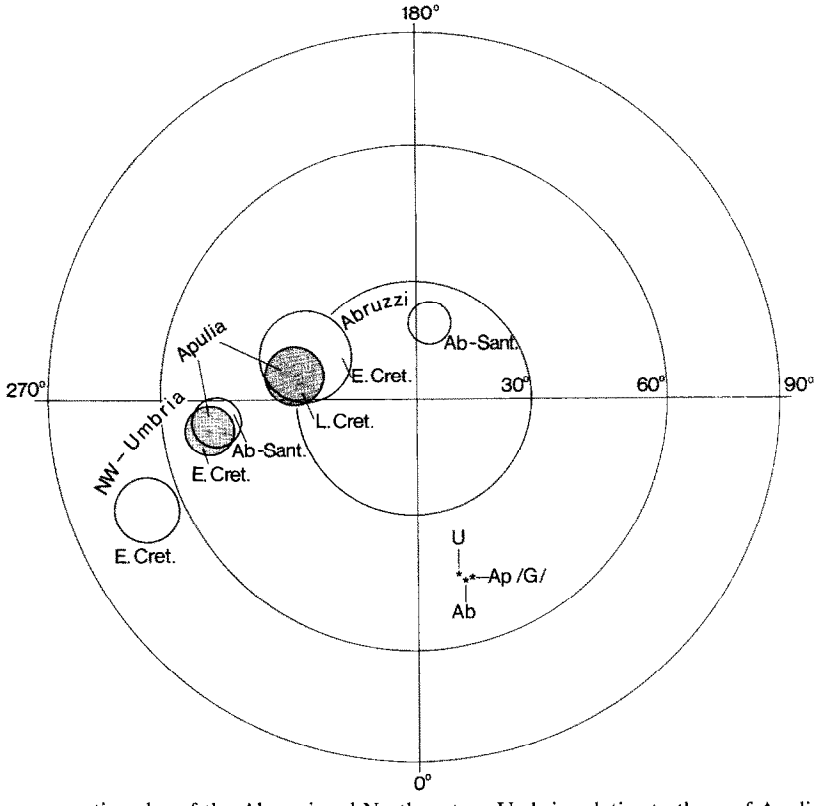


Fig. 5. Cretaceous palaeomagnetic poles of the Abruzzi and Northwestern Umbria relative to those of Apulia. Asterisks show the geographic positions of the Abruzzi (*Ab*), Umbria (*U*) and Apulia (Gargano) (*Ap/G/*).

Apulia in the Tertiary. The counterclockwise rotation of Apulia relative to Africa is probably smaller than the current estimate of 17°.

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References

Celet, P., 1977. The Dinaric and Aegean Arcs: The geology of the Adriatic. In: A.E.M. Nairn, W.H. Kanes and F.G. Stehly (Editors), The Ocean Basins and Margins. Vol. 4, The Eastern Mediterranean. Plenum Press, New York, pp. 215–261.

Channell, J.E.T., 1977. Palaeomagnetism of limestones from the Gargano peninsula (Italy) and the implication of these data. Geophys. J.R. Astron. Soc., 51: 605–616.

Cirilli, S., Márton, P. and Vigliotti, L., 1984. Implications of a combined biostratigraphic and palaeomagnetic study of the Umbrian Maiolica Formation. Earth Planet. Sci. Lett., 69: 203–214.

Colacicchi, R. and Praturlon, A., 1965. Stratigraphical and paleogeographical investigations on the Mesozoic shelf-edge facies in Eastern Marsica (Central Alpines, Italy). Geol. Rom., 4: 89–118.

Consiglio Nazionale delle Richerche, 1981. Carta Tettonica d'Italia. Scala 1:1500000, C.N.R., Rome.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London Ser. A, 217: 295–305.

Hargraves, R.B., 1989. Paleomagnetism of Mesozoic kimberlites in southern Africa and the Cretaceous apparent polar wander curve for Africa. J. Geophys. Res., 94: 1851–1866.

Haq, B.U., Hardenbol, J. and Vail, P.R., 1987. Chronology of fluctuating sea levels since the Triassic. Science, 235: 1156–1167.

Hirt, A.M. and Lowrie, W., 1988. Paleomagnetism of the Umbrian-Marches orogenic belt. Tectonophysics, 146: 91–103.

Irving, E., 1964. Paleomagnetism and its Application to Geological and Geophysical Problems. Wiley, New York.

Jackson, K.C., 1990. A palaeomagnetic study of Apennine thrusts, Italy: Monte Maiella and Monte Raparo. Tectonophysics, 178: 231–240.

Lowrie, W., 1986. Paleomagnetism and the Adriatic promontory: a reappraisal. Tectonics, 5: 797–807.

Lowrie, W. and Heller, F., 1982. Magnetic properties of marine limestones. Rev. Geophys. Space Phys., 30: 171–192.

Lowrie, W., Channel, J.E.T. and Alvarez, W., 1980a. A review of magnetic stratigraphy investigations in Cretaceous pelagic carbonate rocks. J. Geophys. Res., 85: 3597–3605.

Lowrie, W., Alvarez, W., Silva, I.P. and Monechi, S., 1980b. Lower Cretaceous magnetic stratigraphy in Umbrian pelagic carbonate rocks. Geophys. J.R. Astron. Soc., 60: 263–281.

Márton, E. and Nardi, G., 1990. New palaeomagnetic results from Murge (preprint).

McFadden, P.L. and Jones, D.L., 1981. The fold test in palaeomagnetism. Geophys. J.R. Astron. Soc., 67: 53–58.

McFadden, P.L. and Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J.R. Astron. Soc., 67: 19–33.

VandenBerg, J., 1983. Reappraisal of paleomagnetic data from Gargano (South Italy). Tectonophysics, 98: 29–41.

VandenBerg, J., Klootwijk, C.T. and Wonders, A.A.H., 1978. Late Mesozoic and Cenozoic movements of the Italian Peninsula: Further paleomagnetic data from the Umbrian sequence. Geol. Soc. Am. Bull., 89: 133–150.